

Modelling Master Plan — Sirignano-Style Transition Models on the Fannie Mae Panel

Purpose of this folder. The data pipeline (dev/pipeline/) is built: the SSD holds a cleaned loan-month panel with the seven-state target (state, state_next, censored), leakage-safe calendar columns (period_ym, orig_ym), a loan-keyed shard for minibatch randomization, and train-only scaler discipline. This folder specifies the **analysis and modelling** that sits on top of it, replicating Sirignano, Sadhwani & Giesecke (*Deep Learning for Mortgage Risk*, JFEC 2021) on this dataset. Read order: this file → 01_EDA.md → 02_LOAN_LEVEL.md → 03_POOL_LEVEL.md → 04_TASKS.md (sequenced tasks with acceptance criteria). The invariants in the root CLAUDE.md and dev/pipeline/CLAUDE.md continue to bind.

1. Objective in one paragraph

Estimate the **monthly conditional state-transition distribution** of US mortgages — $P(\text{state}_{t+1} \mid \text{state}_t, \text{features}_t)$ over {current, dpd_30, dpd_60, dpd_90plus, foreclosure, REO, prepaid} — with deep neural networks, and quantify the value of nonlinearity by benchmarking against (a) an empirical (frequency-count) transition matrix and (b) multinomial logistic regression, using strictly out-of-sample negative log-likelihood and per-transition AUC. Then roll the fitted loan-level model forward to predict **pool-level** prepayment and delinquency counts over a 12-month horizon, comparing models on predictive accuracy at the pool level — the economically relevant object for MBS analysis.

2. The three phases (sequential)

Phase	Spec	Output
1. EDA & motivation	01_EDA.md	Figures + summary tables establishing data scope, transition base rates, and <i>prima facie</i> nonlinearity/interaction evidence; short memo
2. Loan-level models	02_LOAN_LEVEL.md	Empirical matrix → multinomial logit → deep nets (depth grid, dropout/L2, ensemble); evaluation + robustness; two memos
3. Pool-level models	03_POOL_LEVEL.md	12-month roll-forward of the loan-level model; pool-bucket predictions vs realized; memo

Phase 3 depends on Phase 2's frozen models. Phases 1 and 2 can overlap partially (EDA first — it informs feature transforms).

3. Key design decisions (settled)

- Compute: cloud GPU** (Colab Pro / RunPod / Lambda). Training data is *exported* from the SSD as compact Parquet shards (processed/training/), uploaded once per split design. All training code must run both locally-on-CPU (smoke test on a tiny sample) and on a single cloud GPU. Checkpoints and metrics sync back to models/ on the SSD.
- Split design: rolling expanding-window backtest is the primary design** (per Felipe, June 2026 — supersedes the fixed-split idea in the May-8 notes). One window per **test year** $k = 2015 \dots 2025$ (2025 partial, to the release cutoff) — 11 windows:
 - **Train:** labels $\leq \text{Dec}(k-2)$ — expanding window starting 2000.
 - **Validation:** labels in year $k-1$ (≤ 1 yr, per the design) — used for **early stopping only**.

- **Test:** labels in year k (1 yr ahead).
 - Splits are stated on **label months**; the implementing mask is on the feature month with the strict rule `period_ym < cutoff` (label is one month ahead — 01_SCHEMA.md §6.1).
 - **Hyperparameters/architecture are tuned ONCE** on a designated tuning window ($k = 2015$; spot-check on one later window), then **frozen across all windows** — per-window re-tuning would be a compute explosion and a multiple-testing problem. Per-window val serves only early stopping; scalars/vocabularies/incentive index are re-derived per window from its train slice.
 - **Headline results:** per-test-year NLL/AUC by model, plus a **pooled all-years out-of-sample NLL** — every 2015–2025 test row is predicted exactly once, by a model that never saw it. Regime shifts (COVID 2020–21, the 2022–23 rate spike) appear as test-year effects rather than hinging on one cutoff choice.
3. **Scale: start small, scale up.** All code is developed and validated on a ~5–10 M row stratified sample (fits in RAM, minutes per epoch). Final headline models re-trained on a large export (target ~50–100 M training rows; revisit after seeing throughput). The empirical matrix and logit benchmarks are cheap and can use the *full* training slice via DuckDB streaming regardless.
 4. **Macro covariates remain out of scope** (pipeline decision), with one exception worth its weight: a **rate-incentive proxy derived from the panel itself** — the average Original Interest Rate of loans originated in calendar month t serves as a market-rate index `mkt_rate(t)`; `incentive_t = current_rate - mkt_rate(t)` is the canonical prepayment driver and the paper’s strongest nonlinearity. It uses only contemporaneous data (no leakage). Specified in 01_EDA.md §4 and 02_LOAN_LEVEL.md §2.

4. Known deviations from the paper (state these in the writeup)

- **Dataset:** Fannie Mae public dataset (30-yr FRM, conventional, $LTV \leq 97$) vs the paper’s CoreLogic private-label data. Cleaner credit box, no ARMs/subprime → expect lower base rates of delinquency and different nonlinearity strength.
- **State observability:** in this dataset foreclosure, REO, and prepaid are **terminal** (derived from Zero Balance codes in the loan’s final month); there is no ongoing “in foreclosure” monthly state as in CoreLogic. Origin states are effectively {current, dpd_30, dpd_60, dpd_90plus}; destinations span all seven. The transition matrix is 4×7 , not 7×7 . All models condition on the same origin set, so comparisons remain apples-to-apples.
- **No external macro covariates** (unemployment, HPI by zip). The incentive proxy partially substitutes for the rate channel; the housing/labor channels are absent — a stated limitation and a clean future-work hook.
- **Sample period:** 2000–2025 (vs 1995–2014), so the test regime includes COVID forbearance — note that forbearance-era delinquency codes may behave unusually (check `dq_status` distributions in 2020–21 during EDA).

5. Repository layout this plan creates

```
dev/
├─ model_plan/           # these spec docs
├─ analysis/             # Phase 1 EDA scripts (one figure/table per script, saved to outputs/figures)
├─ model/                # Phases 2–3 code
│   ├── config.py        # split cutoffs, sample sizes, paths (imports pipeline config ROOT)
│   ├── export.py        # build processed/training/<design>/ shards for GPU upload
│   ├── features.py       # feature_spec → model matrix (embeddings map, scaler fit/apply)
│   ├── data.py           # shard-streaming Dataset / DataLoader (epoch = shuffle shards → shuffle within)
│   ├── benchmarks.py     # empirical transition matrix + multinomial logit
│   ├── net.py            # MLP with categorical embeddings
│   ├── train.py          # single-window training loop: early stopping on val NLL, checkpointing, metrics
│   ├── ensemble.py       # K independent fits + probability averaging
│   ├── evaluate.py       # NLL / per-transition AUC / calibration on frozen test slices
│   ├── backtest.py       # THE main harness: loops train.py/ensemble.py over the 11 rolling windows
│   └─ pool.py            # Phase 3: pool construction, 12-month roll-forward, pool metrics
writeup/memos/           # short interim write-ups (markdown), figures referenced from outputs/figures
```

```
models/                # (SSD) one run-folder per fit: config.json, scaler.json, checkpoint, metrics.js
processed/training/    # (SSD) exported training shards per split design
```

6. Non-negotiables (carried over + new)

1. Pipeline invariants hold: stream via DuckDB/Polars, raw/ immutable, require_drive(), idempotent stages.
2. **Temporal masks are strict** (period_ym < cutoff); censored rows (state_next null) are excluded from every likelihood.
3. **The test slice is never downsampled, reweighted, or touched during development.** All evaluation runs on the identical frozen test rows for every model — that's what makes the NLL comparison meaningful.
4. If training data is class-downsampled (thinning current→current), the loss must be **importance-weighted** ($w = 1/\text{keep-prob}$) so the model still estimates true conditional probabilities; reported probabilities must be verified against unsampled base rates (Phase 2 QA).
5. Scalers, embedding vocabularies, and the incentive index are fitted/derived **on the training slice only** (vocab/scaler) or contemporaneously (incentive), persisted under models/, and re-fitted per backtest window.
6. Every model fit writes a self-describing run folder (config, data hash, git commit, metrics) — reproducibility is a dissertation requirement.

7. Backup & recovery

The SSD is a single point of failure only for **small** artifacts; protect those and the rest is rebuildable:

Tier	What	Protection
Code, specs, memos, writeup	this repo	git + GitHub (push after every milestone)
Trained models, run configs, scalars, metrics, QA reports, figures, tables, mkt_rate, export manifests	SSD models/ outputs/ logs/ processed/macro/ + manifests	dev/tools/backup_ssd.sh → ssd_mirror/ on the internal disk (gitignored). Run after every milestone gate — it's in the standing rules.
Parquet lakes (interim/, processed/)	SSD	not backed up — rebuildable from raw/ by re-running the pipeline (~days, no thought)
raw/ (~800 GB)	SSD	not backed up — re-downloadable from Fannie Mae; Stage 1 manifest (in outputs/, hence mirrored) provides checksums/row counts to verify a re-download

Cloud-GPU runs add a free third copy of whatever was uploaded (training shards) and of checkpoints until the instance is torn down — sync checkpoints back to models/ at the end of every GPU session, never leave the only copy on a rented box.

8. Definition of done

- Phase 1: figures + memo motivating nonlinear modelling on this data.
- Phase 2: (a) a tuning-window table replicating the paper's Table 11 structure (in/out-of-sample NLL: empirical matrix, logit, NN depths 1/3/5/7, ensemble); (b) the rolling-backtest results — per-test-year NLL/AUC by model 2015–2025 plus the pooled all-years NLL; (c) calibration plots and robustness (dropout/L2 ablation, ensemble-size curve, seed variance, ranking stability across windows). Memo per milestone.
- Phase 3: pool-level predicted-vs-realized analysis (scatter + R^2 /RMSE by pool bucket scheme) for NN vs logit vs empirical at the 12-month horizon, at a minimum of 3 window anchors (ideally all 11). Memo.